

Assessment 1

Security and Maintenance

Section A: Scenario-Based Concept Application

1. Explain how the installation of Active Directory Domain Services (AD DS) facilitates centralized identity and access control for a small business IT setup.

- Installing Active Directory Domain Services (AD DS) a small business to manage all users, computers, and permissions from one centralized location called a **Domain Controller**.

- ❖ **Benefits:**

- Centralized authentication for all employees.
- Single username and password for accessing company resources.
- Administrators can control permissions from one server.
- Easier management of users, printers, shared folders, and policies.
- Improves security by applying domain-wide rules.

- ❖ **Example:**

- Instead of creating separate accounts on every PC, the administrator creates one domain account in AD DS, and the employee can log in from any company computer.

2. Explain how the implementation of Group Policy Objects (GPOs) enforces robust password complexity and account lockout requirements across a domain.

- Group Policy helps administrators enforce security rules across all domain computers.

- ❖ **Password Complexity Policy:**

A GPO can enforce:

- Minimum password length
- Uppercase/lowercase characters
- Numbers and symbols
- Password expiration

- ❖ **Account Lockout Policy:**

If a user enters the wrong password multiple times:

- The account becomes temporarily locked.
- Prevents brute-force attacks.

- ❖ **Example:**

- Minimum password length: 8 characters
- Lock account after 5 failed attempts for 15 minutes

- This ensures strong authentication security throughout the domain

3. Explain the interaction between NTFS permissions and Shared Folder permissions when managing multi-departmental access to sensitive company data.

❖ When users access shared folders:

- **Share Permissions** control access over the network.
- **NTFS Permissions** control access on the disk.

The **most restrictive permission wins**.

❖ **Multi-Department Example:**

- HR Group → Full Control
- Marketing Group → Read
- Finance Group → No Access

➤ This protects sensitive company files while allowing controlled collaboration.

4. Explain the DHCP DORA process (Discover, Offer, Request, Acknowledge) and how it eliminates manual IP address conflicts for network users.

➤ Dhcp is automatically assigns IP addresses.

Step	Description
Discover	Client searches for DHCP server
Offer	DHCP server offers an IP address
Request	Client requests offered IP
Acknowledge	Server confirms assignment

❖ **Benefits:**

- Prevents duplicate IP conflicts
- Eliminates manual IP configuration
- Saves administrative time
- Automatically assigns:
 - IP address
 - Subnet mask
 - Gateway
 - DNS server

5. Explain the role of DNS Name Resolution in ensuring internal users can consistently access the company's hosted web resources.

- DNS converts domain names into IP addresses.

- ❖ **Benefits:**

- Easier access to resources
 - Faster communication
 - Reliable internal resource connectivity
 - Supports web servers, file servers, and email servers

- Without DNS, users would need to remember IP addresses manually.

6. Explain the technical configuration of a Windows Server Backup Schedule to ensure automated, regular data protection without manual intervention.

- Windows Server Backup allows automated backups.

- ❖ **Configuration Steps:**

1. Install Windows Server Backup Feature
2. Open Windows Server Backup
3. Select "Backup Schedule"
4. Choose:
 - Full Server Backup
 - Custom Backup
5. Set schedule:
 - Daily
 - Weekly
6. Select backup destination:
 - External Drive
 - NAS
 - Another Volume

- ❖ **Benefits:**

- Automatic protection
 - Reduces human error
 - Faster disaster recovery
 - Protects business continuity

Section B: Practical Application Tools Required: VMware/VirtualBox, Windows Server 2022 ISO, Windows 10/11 Client VM.

1. Core Services: Install Windows Server and promote it to a Domain Controller with AD DS and DNS.

❖ **Steps:**

1. Install Windows Server 2022 in VMware/VirtualBox
2. Set static IP address
3. Open:
 - Server Manager
4. Click:
 - Add Roles and Features
5. Install:
 - Active Directory Domain Services
 - DNS Server
6. Promote server to Domain Controller:
 - Manage → Promote this server to a domain controller
7. Create new forest:
 - technova.local
8. Restart server

2. Directory Structure: Create 3 Organizational Units (OUs), 5 User Accounts, and 2 Security Groups via Active Directory Administrative Center.

❖ **Organizational Units:**

- HR
- Marketing
- IT

❖ **User Accounts:**

- user1
- user2
- user3
- user4
- user5

❖ **Security Groups:**

- HR_Group
- Marketing_Group

❖ **Tool:**

- Active Directory Administrative Center

3. Policy Enforcement: Configure a GPO to enforce complex passwords and restrict Control Panel access for specific OUs.

❖ **Password Policy:**

- Open:
 - Group Policy Management
- Navigate:
 - Computer Configuration
 - Policies
 - Windows Settings
 - Security Settings
 - Account Policies
 - Password Policy
- Configure:
 - Minimum 8 characters
 - Complexity Enabled
 - Maximum age 30 days
- Restrict Control Panel:
- Enable:
 - Prohibit access to Control Panel

4. Resource Management: Create a shared folder on an NTFS volume with granular NTFS and Sharing permissions for defined groups.

❖ **Share Permissions:**

- HR_Group → Change
- Marketing_Group → Read

❖ **NTFS Permissions:**

- HR_Group → Modify
- Marketing_Group → Read & Execute

❖ **Steps:**

1. Right-click folder
2. Properties
3. Sharing Tab
4. Advanced Sharing
5. Security Tab

5. Network Automation: Authorize a DHCP scope to automate IP leasing and verify connectivity using ipconfig /renew on a client.

Setting	Value
Start IP	192.168.1.100
End IP	192.168.1.200
Subnet Mask	255.255.255.0
Gateway	192.168.1.1

❖ **Install DHCP:**

- Server Manager → Add Roles → DHCP Server

Section C: Integrated Case Study Scenario: "TechNova," a boutique marketing firm, recently updated its Group Policy Objects (GPOs) to enhance security. Immediately following the update, several employees reported that while they can log into their workstations, they are met with "Access Denied" errors when attempting to reach the centralized Project Files share. The administrator suspects a conflict between the new security templates and existing NTFS permissions, or a failure in GPO replication across the network.

1. Analysis: Identify three specific GPO settings or NTFS conflicts that could cause users to log in successfully but lose access to shared drives.

Cause : 1 deny NTFS permission:

- A new GPO may add users to a restricted group that has:
 - Deny Read
- permission on the shared folder.

Cause 2: User Removed from Security Group

- GPO Restricted Groups or Security Filtering may remove users from:
 - ProjectFiles_Group
- Users can log in but lose file access.

Cause 3: GPO Replication Failure

- One Domain Controller may have old policies while another has updated policies.
- This creates inconsistent authentication and permissions.

2. Troubleshooting: Detail the step-by-step process of using the Resultant Set of Policy (RSOP) or gpresult tool to diagnose this issue.

- ❖ Check:
 - Applied GPOs
 - Denied GPOs
 - Security Group Membership
 - User Rights Assignment

- ❖ **Using RSOP**

Run:

- rsop.msc

Check:

- Security Settings
- Administrative Templates
- Applied Policies
- Identify conflicting GPO settings.

3. Tooling: List the specific CLI commands and GUI consoles you would use to verify if the file server is correctly seeing the user's updated security tokens.

❖ GUI console

Tool	Purpose
Active Directory Users and Computers	Verify group membership
Group Policy Management	Review GPO settings
File Explorer Security Tab	Verify NTFS permissions
Event Viewer	Check authentication errors

❖ CLI Commands

- 1) Check group membership:
 - whoami /groups
- 2) Force policy update:
 - gpupdate /force
- 3) Check network shares:
 - net share
- 4) Check permissions:
 - icacls D:\ProjectFiles

4. Resolution: Propose a final fix to restore access and a preventive strategy, such as A-G-DL-P, to ensure future policy changes do not break permissions.

Step	Meaning
A	Accounts
G	Global Groups
DL	Domain Local Groups
P	Permissions

❖ FIX

1. Verify users belong to correct groups.
2. Remove conflicting Deny permissions.
3. Force replication:
 - repadmin /syncall

4. Run:

- gpupdate /force

Re-login users

- This simplifies permission management and prevents future conflicts.

5. Validation: Explain how to use the net share and icacls commands to confirm that the folder permissions match the intended business logic.

- **Check sheared folder :**

- net share

- Confirms folder is shared correctly.

- Check NTFS Permissions

- icacls D:\ProjectFiles

Symbol Permission

M Modify

RX Read & Execute

- This verifies permissions match business requirements.